Calculative Foundation

Theory Reference: Linear Algebra Concepts for Student Performance Analysis

Duration: 6 Hours | Type: Theory + Practical

This document defines every linear-algebra concept used in the accompanying **Calculative_Foundation.ipynb** notebook, which analyzes a student performance dataset (Math, Science, English, History, Computer scores for 200 students). Each concept includes a definition, the governing formula, and its meaning in the context of the dataset.

Part A — Vector & Matrix Fundamentals

1. Vector

A **vector** is an ordered list of numbers representing a point or quantity with both magnitude and direction in n-dimensional space. In this project, each student's five subject scores form one vector in $\blacksquare^5$:

v = [Math, Science, English, History, Computer]

2. Norm (Vector Length)

A **norm** measures the size/length of a vector.

- **Norm-1 (Manhattan / L1 norm):** sum of absolute values of the components.
- **Norm-2 (Euclidean / L2 norm):** square root of the sum of squared components — the straight-line length of the vector.

$||v||\blacksquare = \sum|v\blacksquare| \qquad ||v||\blacksquare = \sqrt{(\sum v\blacksquare^2)}$

In the dataset, Norm-2 gives an overall 'size' of a student's performance vector, while Norm-1 is closer to a simple total-marks sum.

3. Dot Product & Angle Between Vectors

The **dot product** of two vectors multiplies corresponding components and sums the results. It measures how much two vectors point in the same direction.

$v\blacksquare \cdot v\blacksquare = \sum (v\blacksquare\blacksquare \times v\blacksquare\blacksquare) \qquad \cos(\theta) = (v\blacksquare \cdot v\blacksquare) / (||v\blacksquare|| \times ||v\blacksquare||)$

The resulting **angle** θ between two students' score vectors indicates how similar their performance *pattern* is — a small angle means similar relative strengths and weaknesses across subjects, regardless of their total score.

4. Cross Product

The **cross product** is defined only for 3D vectors and produces a new vector perpendicular to both inputs. Its magnitude equals the area of the parallelogram the two vectors span.

$v\blacksquare \times v\blacksquare = (a\blacksquare b\blacksquare - a\blacksquare b\blacksquare,\ a\blacksquare b\blacksquare - a\blacksquare b\blacksquare,\ a\blacksquare b\blacksquare - a\blacksquare b\blacksquare)$

Applied to the first three subjects (Math, Science, English), it gives a geometric sense of how 'spread apart' two students' profiles are in that 3-subject space.

5. Vector Projection

The **projection** of vector v■ onto v■ is the component of v■ that lies along the direction of v■.

$proj_{v\blacksquare}(v\blacksquare) = [(v\blacksquare \cdot v\blacksquare) / (v\blacksquare \cdot v\blacksquare)] \times v\blacksquare$

This shows how much of one student's performance profile 'points the same way' as another's.

Part B — Matrix Operations

1. Matrix

A **matrix** is a rectangular array of numbers arranged in rows and columns. Stacking several student vectors on top of each other forms a *students* × *subjects* matrix.

2. Matrix Addition & Multiplication

- **Addition:** two matrices of the same shape are added element-by-element.
- **Multiplication (A × B):** the entry in row i, column j of the result is the dot product of row i of A and column j of B. This is only defined when the number of columns of A equals the number of rows of B.

Multiplying a student-score matrix by the transpose of another produces a similarity matrix, where each entry is the dot product between one student from each group.

3. Transpose

The **transpose** of a matrix flips it over its diagonal, turning rows into columns and vice-versa: $(A^T)_{\blacksquare\blacksquare} = A_{\blacksquare\blacksquare}$. Transposing a students×subjects matrix produces a subjects×students matrix.

4. Determinant

The **determinant** is a scalar computed from a square matrix that indicates whether the matrix is invertible (non-zero) or singular (zero), and geometrically represents the scaling factor a matrix applies to volume/area.

5. Inverse

The **inverse** A^{-1} of a square matrix A satisfies $A \times A^{-1} = \mathbf{I}$ (the identity matrix). It exists only when the determinant is non-zero, i.e. the matrix's rows (here, students' score profiles) are linearly independent.

Part C — Linear Transformations & Geometry

1. Line, Plane, and Hyperplane

- **Line (1D):** values from a single subject (e.g. Math) form a one-dimensional line of numbers.
- **Plane (2D):** two subjects together (e.g. Math and Science) place each student as a point on a 2D plane.
- **Hyperplane (n-D, n > 3):** a flat subspace one dimension smaller than the space it lives in. With all 5 subjects, each student is a point in 5D space, which can only be reasoned about mathematically (or reduced in dimension, see the notebook's PCA section) since it cannot be drawn directly.

2. Dimensionality

Dimensionality is the number of independent coordinates needed to specify a point in a space. Moving from 1 subject (line) to 2 (plane) to 3 (3D scatter) to 5 (hyperplane) shows how quickly real datasets become impossible to visualize directly, motivating dimensionality-reduction techniques such as PCA.

Part D — Eigenvalues & Decomposition

1. Covariance Matrix

The **covariance matrix** summarizes how every pair of subjects varies together across all students — diagonal entries are each subject's variance, and off-diagonal entries show how two subjects rise and fall together.

2. Eigenvalues & Eigenvectors

For a square matrix A, an **eigenvector** v is a non-zero vector whose direction is unchanged when A is applied to it — it is only stretched or shrunk by a scalar factor, the **eigenvalue** λ:

$$A v = \lambda v$$

For a covariance matrix, eigenvectors point in the directions of maximum variance in the data, and eigenvalues measure how much variance lies along each direction. The eigenvector with the largest eigenvalue represents the dominant pattern of variation among students (commonly an 'overall ability' axis).

3. Eigen-decomposition

A square matrix can be **decomposed** as $A = V\Lambda V^{-1}$, where V is the matrix of eigenvectors (as columns) and Λ is the diagonal matrix of eigenvalues. This decomposition rewrites the matrix in terms of its fundamental directions of variation and is the mathematical basis of Principal Component Analysis (PCA).

Bonus — Additional Decompositions & Dimensionality Reduction

The project objective also references 'decompositions and dimensionality-reduction techniques'. The notebook extends Part D with the following widely-used methods:

1. LU Decomposition

Factorizes a matrix as A = P L U, where P is a permutation matrix, L is lower-triangular, and U is upper-triangular. It is a standard tool for efficiently solving linear systems and computing determinants/inverses.

2. Singular Value Decomposition (SVD)

Factorizes any matrix (not just square ones) as $A = U\ \Sigma\ V^T$, where U and V are orthogonal matrices and Σ is a diagonal matrix of non-negative **singular values** that rank the 'strength' of each underlying pattern in the data. SVD generalizes eigen-decomposition to non-square matrices and underlies many dimensionality-reduction and recommendation-system algorithms.

3. Principal Component Analysis (PCA)

PCA uses the eigenvectors of the covariance matrix (or equivalently, SVD) to project high-dimensional data onto a smaller number of new axes — the **principal components** — chosen to preserve as much variance as possible. In this project, PCA reduces the 5-subject student data down to 2 dimensions that can be plotted, while retaining most of the meaningful structure (i.e. the separation between Above-Average and Below-Average students).

Summary Table

Concept	Formula	Meaning in this dataset
Norm-1 / Norm-2	$\sum\lvert v\rvert \ / \ \sqrt{(\sum v^2)}$	Overall size of a student's score vector
Dot product & angle	v1·v2 / (‖v1‖ ‖v2‖)	Similarity of two students' score patterns
Cross product	v1 × v2 (3D only)	Spread between two students' 3-subject profiles
Projection	(v1·v2 / v2·v2) v2	Shared direction between two students
Determinant / Inverse	det(A), A■^{1}	Whether student rows are linearly independent
Eigenvalues / vectors	Av = λv	Directions & amount of variance across subjects
PCA	Project onto top eigenvectors	Visualize 5D data in 2D, preserving most variance

Reference companion to **Calculative_Foundation.ipynb** — see the notebook for full worked calculations, results, and interpretations on the student_performance.csv dataset.